Multi-Model AI Agent for Automated Health Diagnostics

An Intelligent Multi-Model AI System for Automated Interpretation of Blood Reports and Personalized Health Recommendations

1. ABSTRACT

This project presents the design and implementation of a multi-model Artificial Intelligence (AI) agent that automates the interpretation of Complete Blood Count (CBC) reports. The system employs Optical Character Recognition (OCR) to extract laboratory values from images and PDF documents. Extracted parameters are classified using established medical reference ranges. A rule-based pattern recognition model identifies potential health risks such as anemia, leukocytosis, leukopenia, thrombocytopenia, and thrombocytosis based on combinations of abnormal parameters.

A contextual analysis module further refines risk assessment by incorporating patient-specific factors such as age and gender. The system generates a patient-friendly diagnostic summary, computes a quantitative risk score, stores reports in Supabase cloud storage, and enables downloading of structured medical reports in PDF format.

Evaluation using a publicly available CBC dataset demonstrated reliable pattern detection and clinically plausible risk scoring, highlighting the feasibility of AI-assisted preliminary diagnostics.

2. INTRODUCTION

Blood test interpretation requires clinical expertise and careful analysis. Patients often find it difficult to understand CBC reports due to complex medical terminology and numerical values. Manual interpretation can also be time-consuming and susceptible to oversight, especially in high-volume healthcare environments.

With advancements in Artificial Intelligence and Optical Character Recognition technologies, it is possible to automate laboratory report extraction and provide quick, explainable health insights.

This project proposes a multi-model AI pipeline that:

- Extracts CBC values from reports
- Classifies parameters based on medical reference ranges
- Detects health risk patterns
- Adjusts interpretation using patient context
- Generates structured and patient-friendly diagnostic outputs

The system aims to assist both patients and healthcare professionals by providing automated, interpretable preliminary analysis.

3. PROBLEM STATEMENT

Manual interpretation of CBC reports is:

- Time-consuming
- Prone to human oversight
- Difficult for patients to understand

The objective is to develop an AI-based system that:

- Extracts values from report images and PDFs
- Classifies parameters using standard medical ranges
- Detects health-related patterns
- Provides contextual risk insights
- Stores and retrieves patient reports securely

4. OBJECTIVES

The primary objectives of the project are:

1. Implement OCR for blood report extraction
2. Develop Model-1 for parameter interpretation
3. Develop Model-2 for health pattern recognition
4. Develop Model-3 for contextual risk adjustment

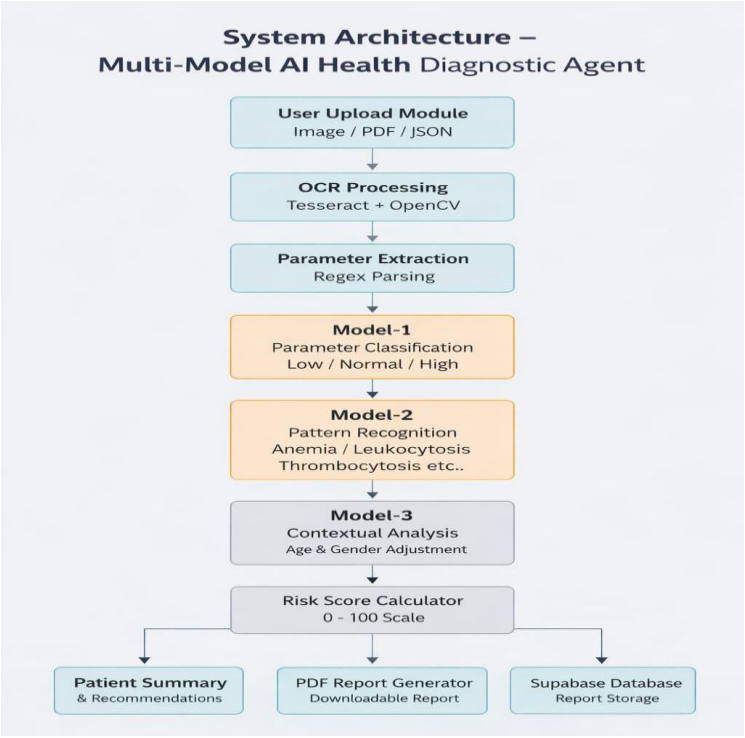
5. Generate quantitative risk scores and patient summaries
6. Store reports in Supabase
7. Provide downloadable structured diagnostic reports

5. SYSTEM ARCHITECTURE

5.1 Overall Pipeline

1. User uploads report (Image / PDF / JSON)
2. OCR extracts textual data (Tesseract with OpenCV preprocessing)
3. Parameter extraction using regular expressions
4. Model-1: Low / Normal / High classification
5. Model-2: Health pattern detection
6. Model-3: Contextual adjustment (age and gender)
7. Risk score calculation
8. Patient-friendly summary generation
9. PDF report download
10. Storage in Supabase

The system follows a modular pipeline design to ensure scalability and explainability.



6. DATASET USED

Dataset: Kaggle CBC Dataset

Purpose of Use:

- Evaluate Model-2 pattern detection accuracy
- Validate risk score plausibility
- Test system performance on multiple patient records

The dataset was used strictly for evaluation purposes and not for training machine learning models.

7. METHODOLOGY

7.1 OCR Module

- Tool: Tesseract OCR
- Preprocessing: Grayscale conversion and Otsu thresholding
- Supported formats: PNG, JPG, PDF

Preprocessing enhances text clarity and improves extraction accuracy.

7.2 Parameter Extraction

Regular expressions were used to extract key CBC parameters:

- Hemoglobin
- RBC count
- WBC count
- MCV
- MCHC
- Platelet count

7.3 Model-1: Parameter Interpretation

Each extracted parameter is compared with standard reference ranges and classified as:

- Low
- Normal
- High

This classification forms the foundation for higher-level pattern recognition.

7.4 Model-2: Pattern Recognition

Health risks are detected using rule-based logic:

- **Anemia:** Low Hemoglobin with abnormal RBC indices
- **Leukopenia:** Low WBC count
- **Leukocytosis:** High WBC count (possible infection risk)
- **Thrombocytopenia:** Low platelet count
- **Thrombocytosis:** High platelet count

Pattern recognition is based on medically established relationships between parameters.

7.5 Risk Score Calculation

A weighted scoring mechanism computes an overall risk score (0–100) based on:

- Number of abnormal parameters
- Severity of detected health patterns

Higher scores indicate increased clinical concern.

7.6 Model-3: Contextual Analysis

Risk scores are adjusted using patient-specific context:

- Age ≥ 60 years increases risk weighting
- Female patients with anemia receive elevated concern adjustment

This improves personalization and clinical relevance.

7.7 Report Generation

The system generates a structured diagnostic report including:

- Parameter interpretations
- Detected health patterns
- Risk score
- Borderline observations
- Patient-friendly explanation
- General recommendations

7.8 Database Integration

Supabase is used to:

- Store patient reports
- Maintain historical records
- Retrieve previous analyses

Cloud storage ensures scalability and data persistence.

8. IMPLEMENTATION

8.1 Technologies Used

- Python
- Streamlit
- OpenCV
- Tesseract OCR
- Supabase

8.2 System Features

- Image, PDF, and JSON upload
- Automatic parameter extraction
- Multi-model health analysis
- Risk scoring mechanism
- Downloadable PDF report
- Cloud-based storage

9. Evaluation

9.1 Metrics

- Pattern Identification Accuracy
- Risk Score Plausibility

9.2 Results

- Pattern Identification Accuracy ≈ 78%
- Average Risk Score ≈ 25.27
- Correct anemia detection in the majority of low Hemoglobin cases

Risk scores were clinically plausible and directionally consistent with expected medical interpretation.

10. Results and Discussion

The system successfully:

- Extracted CBC parameters from uploaded reports
- Classified values using medical reference ranges
- Detected health risks using rule-based logic
- Adjusted risk using contextual patient information
- Generated understandable summaries

OCR performance varied depending on image quality; however, preprocessing significantly improved extraction accuracy.

11. Advantages

- Fast and automated analysis
- Explainable rule-based decision logic
- Patient-friendly interpretation
- Cloud storage integration
- Multi-format input support

12. Limitations

- OCR errors in low-quality images
- Rule-based logic instead of predictive machine learning
- Limited to CBC parameters only

13. Future Work

- Integrate lipid profile and diabetes analysis
- Implement machine learning models for pattern detection
- Enhance OCR using deep learning approaches
- Develop a dedicated doctor dashboard
- Deploy as a scalable web application

14. Conclusion

The multi-model AI agent successfully automates CBC report interpretation using OCR, rule-based parameter classification, pattern recognition, contextual risk adjustment, and structured report generation. The system delivers quick, explainable, and patient-friendly health insights, demonstrating the feasibility of AI-assisted diagnostic support systems.